Consider the following for the next **two (02)** items that follow :

An unbiased coin is tossed n times. The probability of getting at least one tail is p and the probability of at least two tails is q and $p - q = \frac{5}{32}$.

115. What is the value of n ?

(a) 4

(b) 5

(c) 6

(d) 7

116. What is the value of $p + q$?

(a) $\frac{57}{32}$

(b) $\frac{53}{32}$

(c) $\frac{51}{32}$

(d) 1

Consider the following for the next **two (02)** items that follow :

x_i	1	2	3	...	n
f_i	1	2^{-1}	2^{-2}	...	$2^{-(n-1)}$

117. What is $\sum_i^n x_i f_i$ equal to ?

(a) $\frac{2^{n+1} - n + 2}{2^{n-1}}$

(b) $\frac{2^{n+1} - n - 2}{2^{n-1}}$

(c) $\frac{2^{n+1} + n + 2}{2^{n-1}}$

(d) $\frac{2^{n+1} - n - 2}{2^n}$

118. What is the mean of the distribution ?

(a) $\frac{2^{n+1} - n + 2}{2^n - 1}$

(b) $\frac{2^{n+1} - n - 2}{2^{n-1}}$

(c) $\frac{2^{n+1} - n - 2}{2^n - 1}$

(d) $\frac{2^{n+1} - n + 2}{2^n}$

Consider the following for the next **two (02)** items that follow :

The marks obtained by 10 students in a Statistics test are 24, 47, 18, 32, 19, 15, 21, 35, 50 and 41.

119. What is the mean deviation of the largest five observations ?

(a) 4·8

(b) 5·5

(c) 6

(d) 7·5

120. What is the variance of the largest five observations ?

(a) 14·6

(b) 21·8

(c) 25·2

(d) 46·8

कच्चे काम के लिए जगह